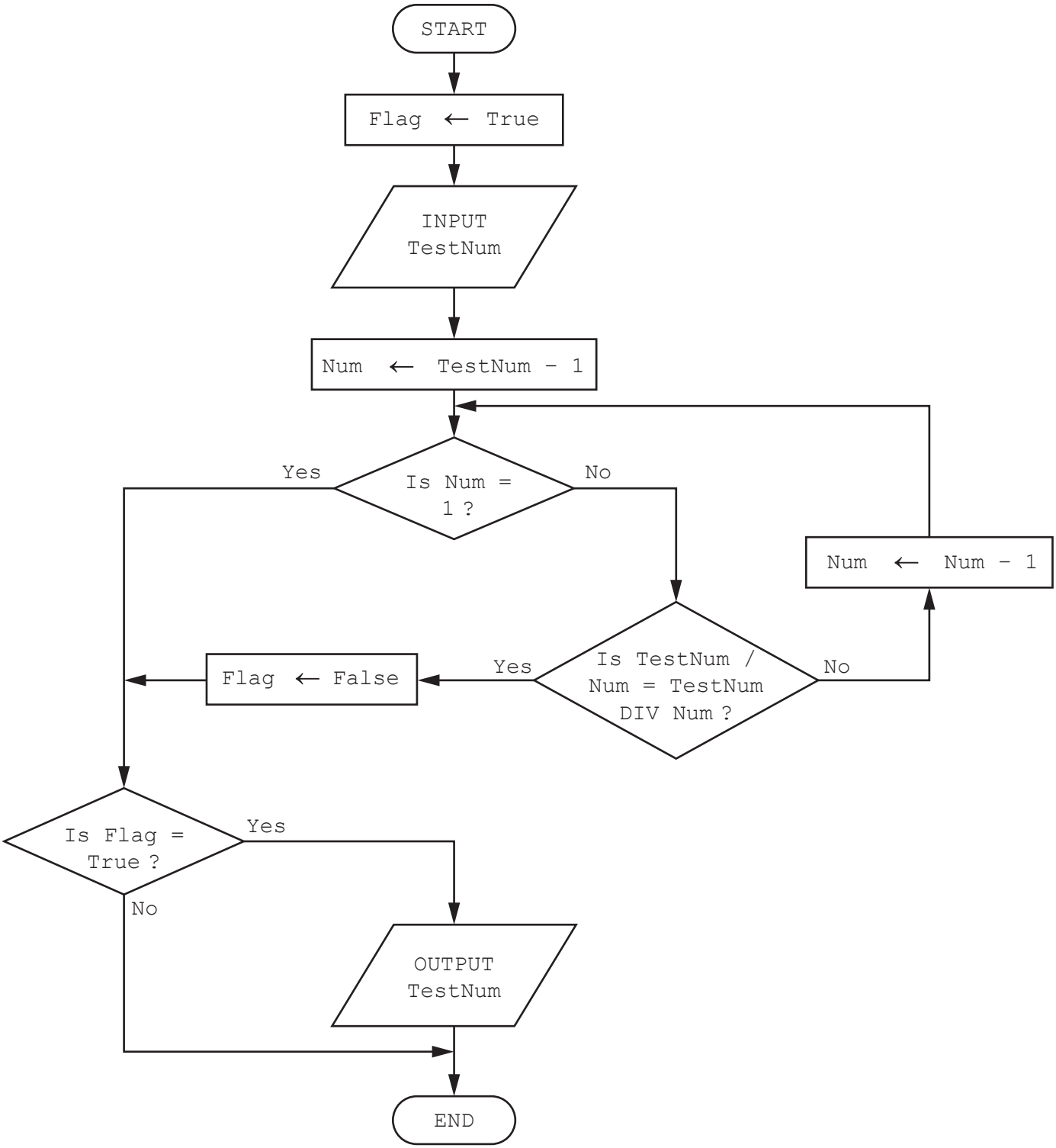
5 The flowchart performs a mathematical process on a number input called `TestNum`

`DIV` is used to represent **integer division** e.g. `7 DIV 3 = 2`



(a) Complete the trace table for the input data: **7**

Flag	TestNum	Num	OUTPUT

[2]

(b) Complete the trace table for the input data: **6**

Flag	TestNum	Num	OUTPUT

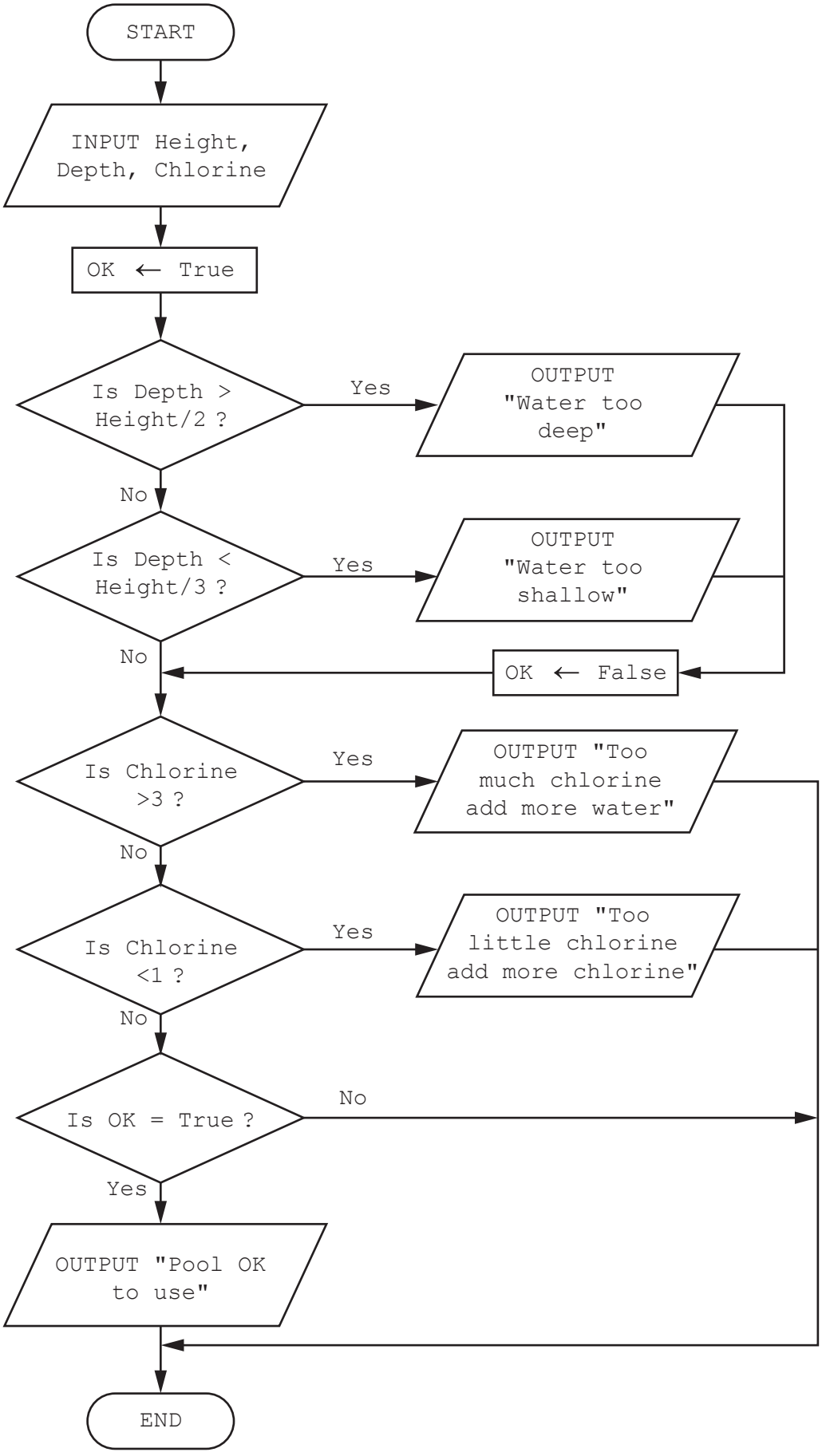
[2]

(c) State the purpose of the algorithm in the flowchart.

.....

.....[1]

5 The flowchart checks the level of chlorine and the depth of water compared to the height of the swimming pool. Error messages are output if a problem is found.



(a) Complete the trace tables for each set of input data.

Input data: **6**, **2.5**, **2**

Height	Depth	Chlorine	OK	OUTPUT

Input data: **4**, **3**, **1.5**

Height	Depth	Chlorine	OK	OUTPUT

Input data: **6**, **3.5**, **4**

Height	Depth	Chlorine	OK	OUTPUT

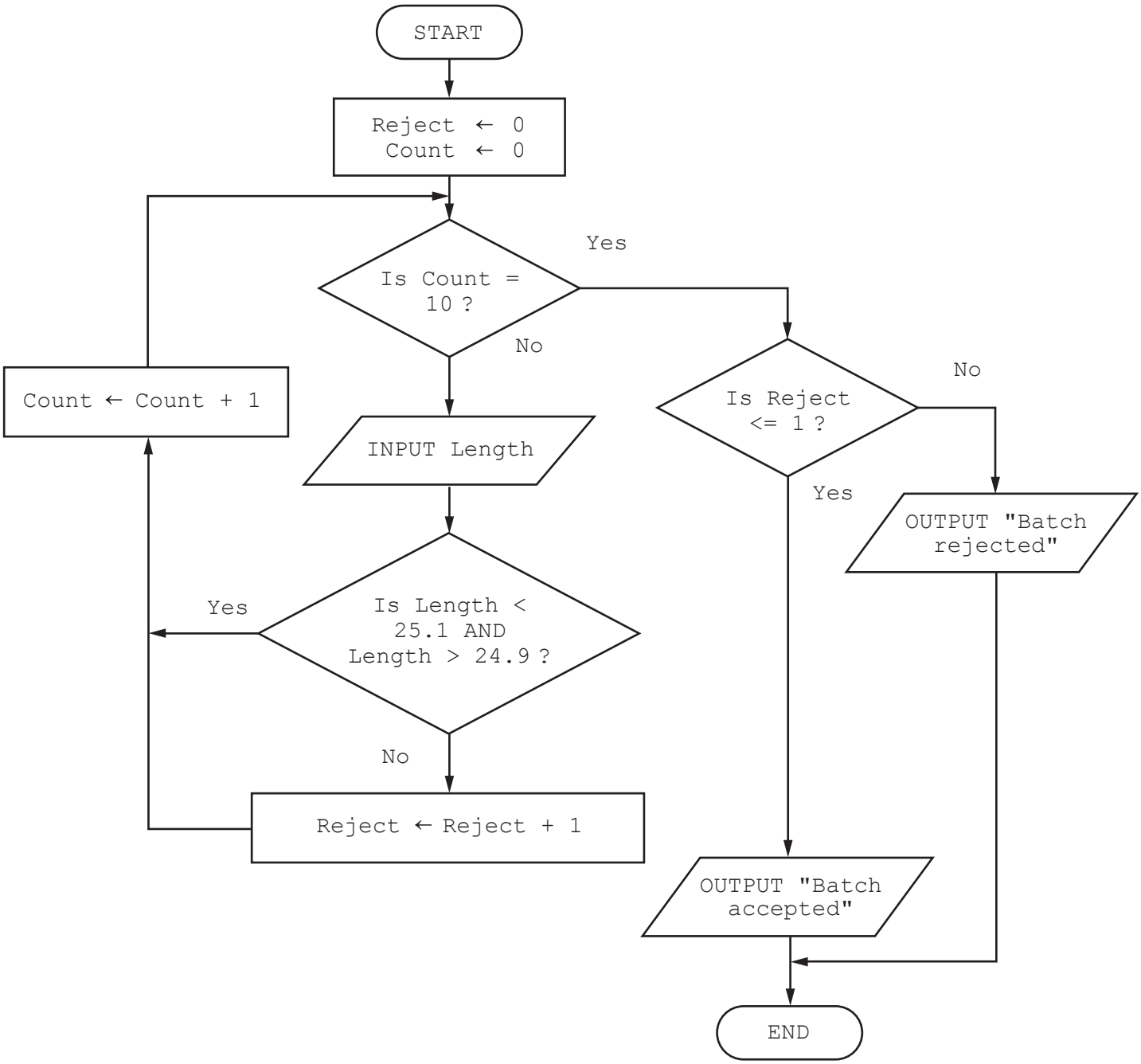
[6]

(b) Identify a problem with the algorithm that the flowchart represents.

.....

..... [1]

3 **(a)** The flowchart checks the lengths of a batch of 10 ropes. For the batch to be accepted 90% of the lengths need to be between 24.9 and 25.1 metres.



Complete the trace table for the input data:

24.88, 25.01, 24.98, 25.00, 25.05, 24.99, 24.97, 25.04, 25.19, 25.07

Reject	Count	Length	OUTPUT

[4]

(b) (i) It has been decided to only reject batches of rope that contain ropes that are too short.

State the change required to the algorithm.

.....

..... [1]

(ii) Explain how the algorithm to reject batches could be improved to make it more effective.

.....

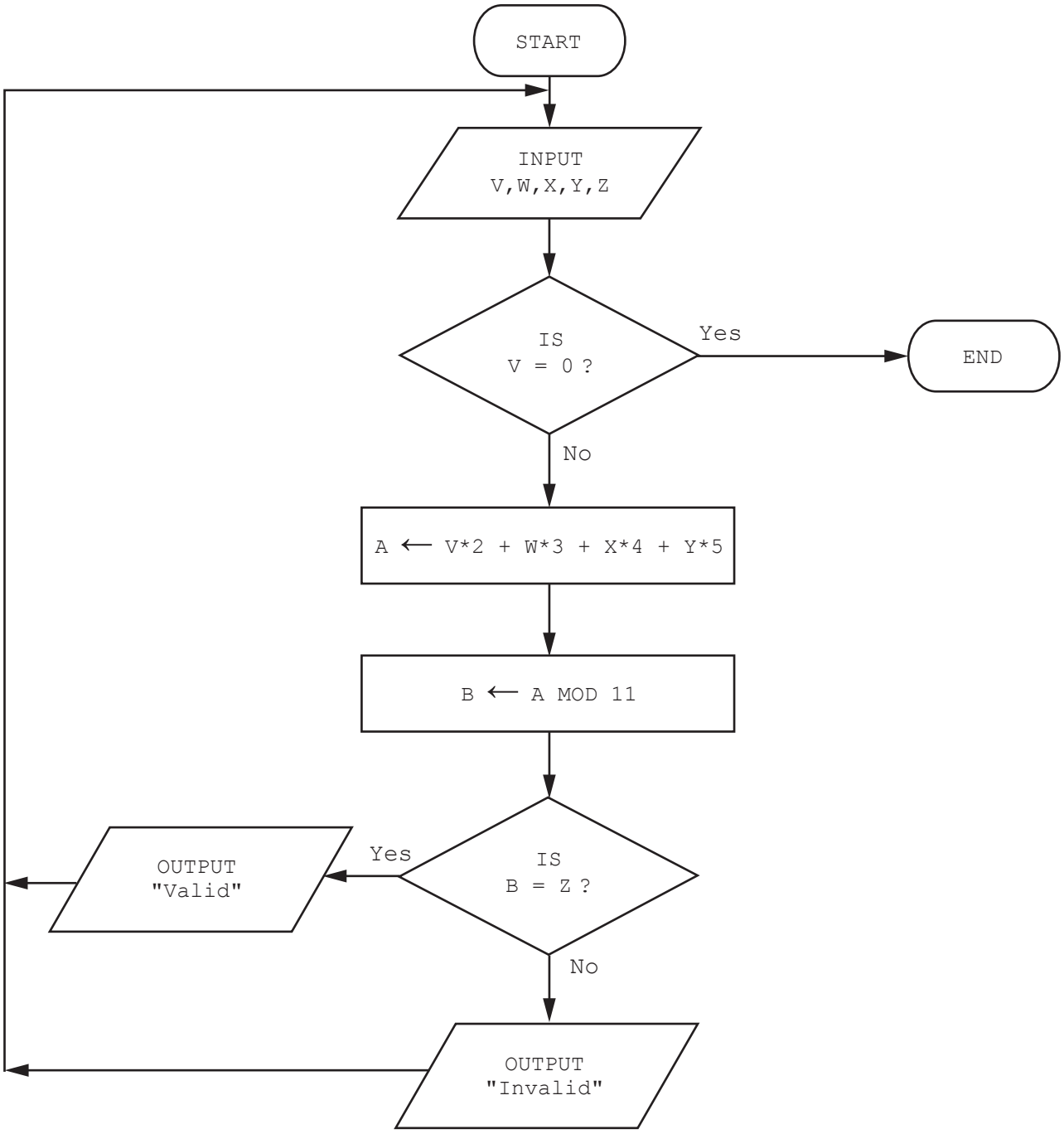
.....

.....

..... [2]

4 This flowchart inputs five numbers and performs a calculation.

The predefined function MOD finds the remainder from integer division for example `R ← 25 MOD 11` gives `R` a value of `3`



(a) Complete the trace table for this set of input data:
5, 4, 6, 2, 1, 9, 3, 2, 1, 6, 7, 6, 1, 5, 1, 0, 0, 0, 0, 0

V	W	X	Y	Z	A	B	OUTPUT

[4]

(b) Describe the purpose of this flowchart.

.....

.....

.....

..... [2]